

EE5750 Quantum Error Correction

Homework 5

Due: 03 Sep 2026

As part of the Homework assessment, provide solutions to only the questions marked as **(*Homework*)**. The rest of the questions are practice problems.

1. **(*Homework*)** (5 marks) *Computing Schmidt decompositions.*

For each state below, find a Schmidt decomposition, state the Schmidt rank r and the Schmidt coefficients μ_k , and say whether the state is entangled. You can use a computer program to find the SVD if you have to.

- (a) $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$ on $\mathbb{C}^2 \otimes \mathbb{C}^2$
- (b) $\frac{1}{\sqrt{3}}(|00\rangle + |11\rangle + |21\rangle)$ on $\mathbb{C}^3 \otimes \mathbb{C}^2$

2. *Superposition versus Mixture*

We saw in Lecture 10 that unitary gates are reversible quantum operations. We do not expect a reversible operation to introduce randomness into a system, and similarly, if there is randomness pre-existing in a system we do not expect a reversible operation to eradicate this randomness. This is true in the quantum case as well: a unitary operator does not introduce new randomness or eradicate existing randomness in a system.

- (a) Consider the computational basis states $|0\rangle$ and $|1\rangle$ in $\mathbb{C}^2$. Would you consider these states as random or definite/non-random?
- (b) Consider the qubit state $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$. Is there a unitary operation that maps the state $|0\rangle$ to $|\psi\rangle$? Identify one such unitary operator. Is there a unitary that maps $|\psi\rangle$ to $|0\rangle$?
- (c) $|\psi\rangle$ and $|0\rangle$ can be transformed into each other through reversible operations. Would you consider $|\psi\rangle$ to be random or non-random?
- (d) Let ρ be a qubit mixed state (not pure). What is the rank of this operator? Is there a unitary gate that transforms this state into a classical state (diagonal density operator) $\rho_{\text{classical}}$? Is $\rho_{\text{classical}}$ a constant state or a classical random variable? Since ρ and $\rho_{\text{classical}}$ are related through a reversible operation, would you consider ρ to be random or non-random?

3. **(*Homework*)** (20 marks) *A secure communications scenario: classical versus quantum*

Alice would like to send Bob two classical bits of information. There is an eavesdropper (Eve) who can access one of the two systems Alice sends to Bob: if Alice sends Bob two classical bits, Eve can access one of these two bits; if Alice sends Bob two qubits, Eve can access one of the two qubits. We will look at a quantum encoding strategy in which no information leaks to Eve.

Quantum encoding by Alice. Alice starts with two physical qubits A and B and prepares them in the EPR state $|\text{EPR}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. She has two classical bits of information b_1, b_2 , which she encodes into the qubits as follows:

$$|\psi\rangle_{AB} = (X^{b_1} \otimes I)(Z^{b_2} \otimes I)|\text{EPR}\rangle.$$

That is, Alice applies Z to qubit A if $b_2 = 1$, and then applies X to qubit A if $b_1 = 1$. She then sends the registers A and B to Bob. Eve might get access to exactly one of these registers. We now analyze two things: decoding by Bob (assuming Eve does not tamper) and security against Eve (in case Eve tampers to extract information about b_1, b_2).

- (a) What are the four possible values of the encoded state $|\psi\rangle_{AB}$ corresponding to the four possibilities for (b_1, b_2) ?
- (b) Verify that these four encoded states are orthonormal in $\mathbb{C}^2 \otimes \mathbb{C}^2$.
- (c) Identify a unitary gate $U \in \mathbb{C}^{4 \times 4}$ that maps these four states to the computational basis states $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ (in some order).

Remark: Observe that, when Eve does not tamper with A or B , Bob can retrieve the information (b_1, b_2) from $|\psi\rangle_{AB}$ using U and a classical measurement. Bob applies the gate U to $|\psi\rangle_{AB}$; he then holds one of the four computational basis states, and a classical measurement tells him which one occurred.

Now let us consider the situation where Eve gets access to one of the two qubits A or B .

- (d) For each of the four possible values of the encoded state $|\psi\rangle_{AB}$ and each of the two possible interceptions (Eve accesses A , or Eve accesses B), find the state of the qubit that Eve intercepts (eight cases in all). Is there any variation in the state of the intercepted qubit across these eight cases?
- (e) How can you argue that Eve can gather absolutely no information about (b_1, b_2) ? Does *indistinguishability of preparations* help?

Now consider the classical case. Suppose Alice uses two classical bits, stored in registers A and B (the names A, B are reused here for this separate classical scenario), to encode the information about (b_1, b_2) , and Eve can intercept one of the two classical bits A or B . Assume Alice's classical encoding is a deterministic function $(b_1, b_2) \mapsto (A, B)$.

- (f) For successful decoding by Bob (in case Eve does not tamper), show that the four possible values of (A, B) must be distinct.
- (g) Now argue that no matter which classical bit Eve intercepts, she can gather *some* information about (b_1, b_2) .

4. (*Homework*) (10 marks) *Quantum circuits for entangled states*

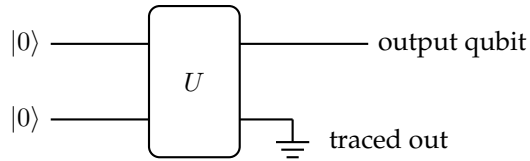
You must design quantum circuits to prepare desired pure states (on a composite system of multiple qubits). In each case sketch the circuit and argue that the circuit functions as desired. The initial states of all the qubits must be $|0\rangle$. You are allowed to use only the following gates: Paulis, Hadamard, CNOT, Toffoli (see HW2, question 3).

- (a) $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$
- (b) $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$
- (c) $\frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$
- (d) $\frac{1}{2}(|000\rangle + |010\rangle + |100\rangle + |111\rangle)$

5. (*Homework*) (5 marks) *Quantum preparation of classical randomness*

In this question, you must use pure states, unitary operations and tracing out to generate classical randomness. In each case you can use a composite system consisting of one or more qubits. All qubits must be initialized to $|0\rangle$.

In a quantum circuit, tracing out (discarding) a qubit is drawn as a wire that ends in the “ground” symbol shown below; the state of the remaining qubits is the partial trace of the circuit output over every wire that ends this way.



In each case, you must draw the quantum circuit for your preparation — using the trace-out notation above — and analytically show that the circuit output holds the desired classical state.

You are allowed to use only the following gates: Paulis, Hadamard, CNOT, Toffoli.

- (a) A classical random variable that can take two possible values 0 and 1 with probabilities $1/4$ and $3/4$, respectively.
 - (b) Two classical bits A and B that are each uniformly random but always equal: the pair (A, B) takes the value $(0, 0)$ with probability $1/2$ and the value $(1, 1)$ with probability $1/2$, and never takes the values $(0, 1)$ or $(1, 0)$.
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